Learning of Ratio and Proportion Problem-Solving Using Schema-Based Instruction: A Replication Study

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Driven to Discover℠

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PURPOSE & FOCUS OF STUDY

The purpose of this study was to replicate and extend the study of the SBI intervention conducted by Jitendra et al. (2015) that demonstrated impact in proportional problem solving for a homogeneous sample of Upper Midwest students and teachers who were predominantly White and middle class. While the importance of replication in enhancing external validity is a staple of educational methods textbooks, replications are rare in practice (Duncan, Engel, Claessens, & Dowsett, 2015; Yong, 2012) but badly needed (Makel & Plucker, 2014).

We chose to replicate this study in two geographically diverse locations (Northern Rocky Mountains and Southeast U.S.) that reflect a diversity of students and teachers across the factors of race, socio-economic status, and the percentage of English language learners.

RESEARCH QUESTIONS

1. What are the effects of the SBI intervention compared to business-as-usual instruction on seventh-grade students' proportional problem solving at immediate posttest and after 11 weeks?
2. What are the effects of the SBI intervention compared to business-as-usual instruction on seventh-grade students' general mathematical problem solving?
3. Does the treatment moderate 7th-grade students' learning trajectory over time?

PARTICIPANTS

Students from 59 seventh-grade classrooms (N = 1,511) and their teachers (J = 59) from 36 middle schools in five school districts participated in the study.

Variable	Level	Percent
Gender	Male	49.4
	Female	49.3
	Missing	1.3
Race	White	55.7
	Black	7.9
	Hispanic	25.9
	Asian	4.7
	Multiracial	3.2
	Missing	2.5
Eligible for Free or Reduced Price Lunch	Yes	23.4
	No	10.1
	Missing*	66.5
English Language Learner Status	Yes	8.7
	No	90.0
	Missing	1.3
Students Receiving Special Education Services	Yes	8.9
	No	89.9
	Missing	1.2

**Note. FRL status at the student level was unavailable for two of the participating school districts (17 schools, 32 classrooms). The students are used in the main analysis with the FRL variable at the classroom level. The school level FRL percentage is used as the value for FRL in these classrooms.*

RESEARCH DESIGN

We used a randomized cluster design with longitudinal data. For each of the 59 teachers, one class of students was randomly selected to participate in the study. Each of the 59 teachers and their participating class was randomly assigned to one of two conditions: treatment or control.

INTERVENTION

SBI was implemented in treatment classrooms, five days a week for 45-50 min over 6 weeks. In the same time period, students in the control condition were taught the same topics (e.g., rates/proportion, scale drawing, percent) using instructional practices specified in their mathematics textbook. Within SBI, teachers used the following four instructional practices of (a) priming the mathematical structure of problems, (b) visually mapping information in the problem using schematic diagrams (see figure below), (c) providing explicit instruction on a problem-solving heuristic, and (d) developing procedural flexibility.

Problem Type	*Example of Problem*
Ratio Compared: 75 Main line lunches / 100 Salad bars lunches = x Base	On Thursday, the cafeteria at Osseo Middle School sold: 42 smoothies, 75 main line lunches, 80 cookies, 51 bags of chips, 100 salad bar lunches, and 26 breakfast bars. What is the ratio of the number of main line lunches sold to the number of salad bar lunches sold on Thursday?
Proportion *If*: \$50 → 27 British pounds; *Then*: \$75 → x British pounds	The Frank family from Minnesota, USA, is going to Britain for their summer vacation. They exchanged \$50 for 27 British pounds. At that exchange rate, how many British pounds could they get for \$75?
Percent of change Change \$8 / Original \$200 = 4/100 \$200 Original & \$8 Change = \$$x$ New	In New York, where the sales tax is 4%, what would be the total cost with tax of a leather coat that costs \$200?

FIDELITY OF IMPLEMENTATION

Fidelity of Implementation (FOI) – Two measures assessed procedural fidelity and adherence to the SBI intervention (e.g., identifies problem type, estimates solution, uses multiple solution strategies, checks solution) as well as quality of instruction in classrooms (e.g., sets lesson purpose, provides lesson closure) .

Teachers' mean total scores (out of a possible 21 points) on the fidelity checklist were 15.71 (SD = 2.42) for SBI teachers and 10.16 (SD = 2.70) for control teachers. Results indicated statistically significant differences between the treatment and control groups on four of the seven items and the total score for fidelity-of-implementation items. With regard to quality of instruction, as expected, there were no differences between treatment (M = 8.75) and control (M = 8.56) groups.

MEASURES

Proportional Problem-Solving Test – 22 multiple-choice items and four short-answer conceptual items that assessed students' knowledge of ratio, proportion, and percent. Reliability estimates for the PPS measure at pretest, posttest, and retention test were 0.77, 0.84, and 0.84, respectively.

Process and Applications Subtest of the Group Mathematics Assessment and Diagnostic Evaluation (GMADE) – 30 multiple-choice items measuring students' ability to comprehend mathematical language and concepts and apply relevant operations to solve word problems across multiple content areas (e.g., algebra, geometry). Pretest and posttest reliability estimates of the GMADE were 0.69 and 0.76, respectively

RESULTS

Results of a series of multilevel (i.e., two-level, students within clusters/teachers) models for cross-sectional data with covariates at both levels indicated statistically significant differences favoring SBI on the PPS posttest (γ = 1.97, $t(50)$ = 4.38, p = <.001, g = 0.48), the retention test (γ = 1.16, $t(50)$ = 2.4, p = .020, g = 0.28), and the GMADE posttest (γ = 0.99, $t(50)$ = 2.27, p = .028, g = 0.25).

In addition, there was evidence that linear slopes over time for PPS varied across students and that this variation was moderated by treatment (γ = .71, $t(1,284)$ = 5.76, p = <.001). These findings suggested that the treatment group was associated with a larger (positive) linear slope over time compared to the control group.

	SBI			Control			Total		
	n	*M*	*SD*	*n*	*M*	*SD*	*n*	*M*	*SD*
PPS pretest	808	10.93	4.97	633	10.82	4.55	1441	10.88	4.79
PPS posttest	805	15.42	6.15	625	12.55	5.18	1430	14.16	5.91
PPS delayed	769	14.26	6.06	580	12.53	5.40	1349	13.52	5.84
GMADE pretest	816	11.68	4.35	625	11.89	4.06	1441	11.77	4.22
GMADE posttest	796	13.85	4.87	598	12.60	4.62	1394	13.31	4.80

Note. SBI = schema-based instruction; PPS = proportional problem solving; GMADE = Group Mathematics Assessment and Diagnostic Evaluation. All test statistics are based on the total number of items correct.

CONCLUSIONS

The preliminary results of this study are generally consistent with previous findings (Jitendra et al., 2009, 2011, 2013, 2015) that SBI enhances proportional problem solving. In both the present study and the original Jitendra et al. (2015) study, the effect sizes on the PPS posttest (g = 0.48 vs. 0.46) and delayed posttest (g = 0.28 vs. 0.32) were about the same, with students in SBI classrooms learning the content more effectively than control students and sustaining the learning over time. We are not only encouraged that the effects in Jitendra et al. (2015) with a homogenous group of students appear to hold in the current study with a more geographically and demographically diverse sample of students, but also by the superiority of SBI over instruction in the control condition in facilitating transfer to solve novel word problems.

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